BIG DATA PROJECT

WASIN S.



MOVIE TICKET BOOKING SYSTEM

OBJECTIVE

1. uses MongoDB Atlas as the data storage

2. functional website that includes information like, avaliable movies, showtimes, seat numbers, and booking history

3. users are able to insert their information into the website which later transfer to MongoDB Atlas



PROJECT OBJECTIVE

LANGUAGE USED:

Python
- backend coding

HTML
- website design

JavaScript
- interactive website

CSS
- styling website









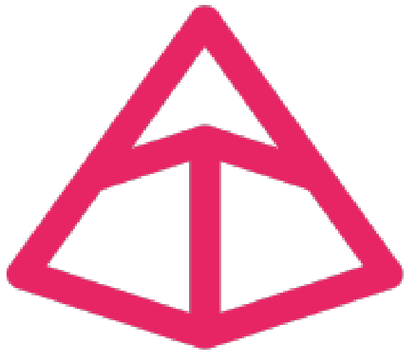
PROJECT MODULES

MODULE USED

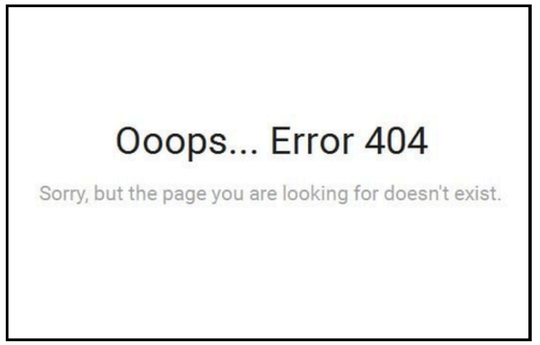
MongoClient:	**connector for VS Code and MongoDB**
FastAPI:	**web framework to build APIs**
HTTPException:	**raise standard HTTP errors**
BaseModel:	**define structures for data**
ObjectId:	**unique id used for MongoDB documents**
CORSMiddleware:	**uses backend from various origins**











BACKEND

03

database.py

- a connector for VS Code and MongoDB Atlas

movies.py

- insert movies into MongoDB Atlas

users.py

- create new user and detect already exist users

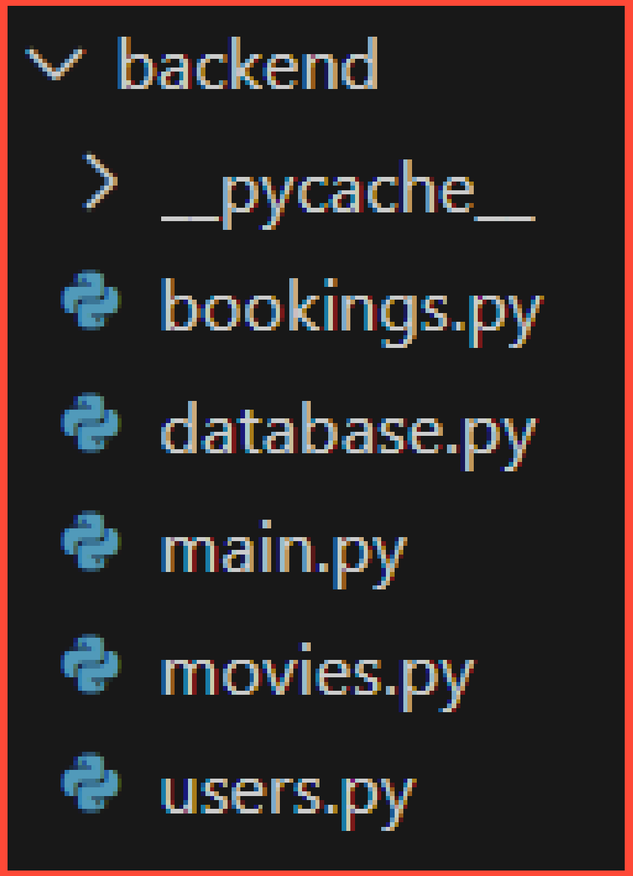
booking.py

- create new booking and detect booked seats

main.py

- main backend server, receives request from frontend and then interacts with the database

PROJECT STRUCTURE



FRONTEND 04

index.html

- main interactive website, used for booking movies, see showtimes, and creating account

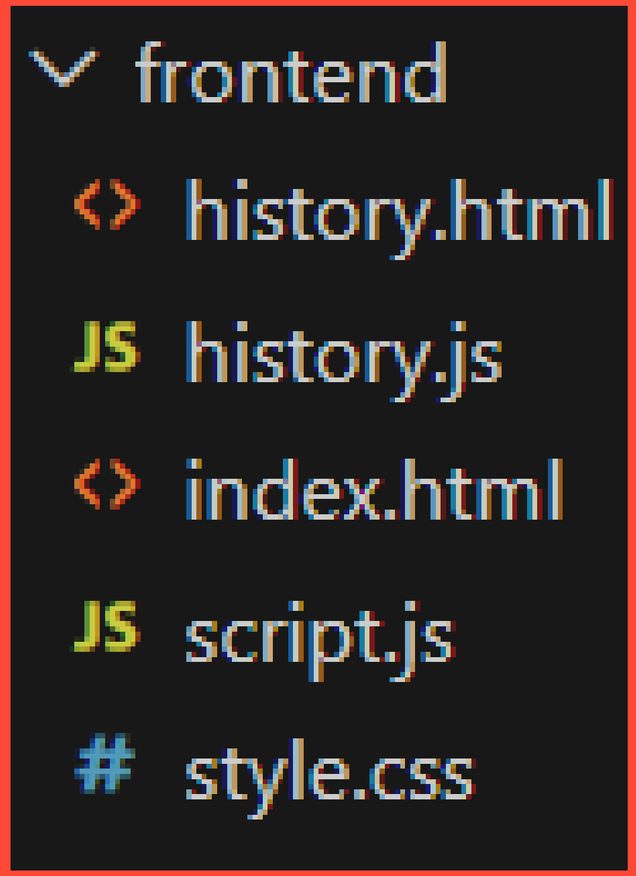
script.js

- handles user interactions, send API request to the backend, process the received data in the frontend, and then display it in the index.html

history.html

- the booking history page, an extention from the main website

PROJECT STRUCTURE



FRONTEND 04

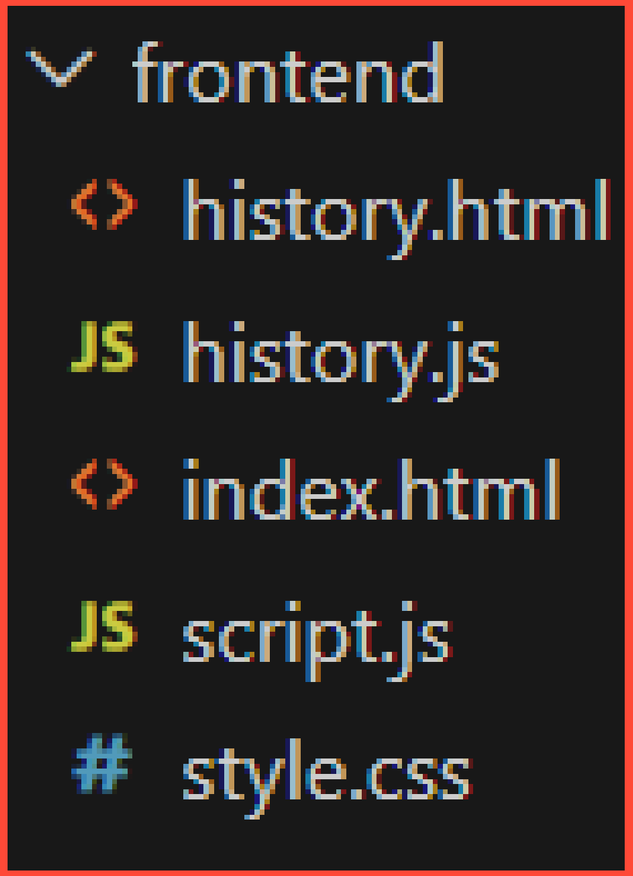
history.js

- same as script.js, but only request and process user's past bookings data and display it

style.css

- giving the website a better look, changing background and font colors, font style and size, text alignment, etc.

PROJECT STRUCTURE



backend > main.py > book_ticket

```python
from fastapi import FastAPI, HTTPException
from users import create_user
from bookings import create_booking
from database import movies_collection, bookings_collectio
from pydantic import BaseModel
from bson import ObjectId
from fastapi.middleware.cors import CORSMiddleware

app = FastAPI()

@app.get("/")
def root():
    return {"message": "Movie Booking API is running."}

# enable CORS
app.add_middleware(
    CORSMiddleware,
    allow_origins=["*"],
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)
```

frontend > script.js > bookTicket

```javascript
let moviesData = [];
let seatRequestVersion = 0;
let selectedSeat = null;

// load movies from backend
async function loadMovies() {
    const response = await fetch("h
    const movies = await response.j

    moviesData = movies;

    const movieSelect = document.ge

    // add placeholder option
    const placeholder = document.cr
    placeholder.value = "";
    placeholder.text = "--- Select
    placeholder.disabled = true;
    placeholder.selected = true;
    movieSelect.appendChild(placeho

    // add real movies
    movies.forEach(movie => {
        const option = document.cre
```

DEMO

05

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THANK YOU!

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